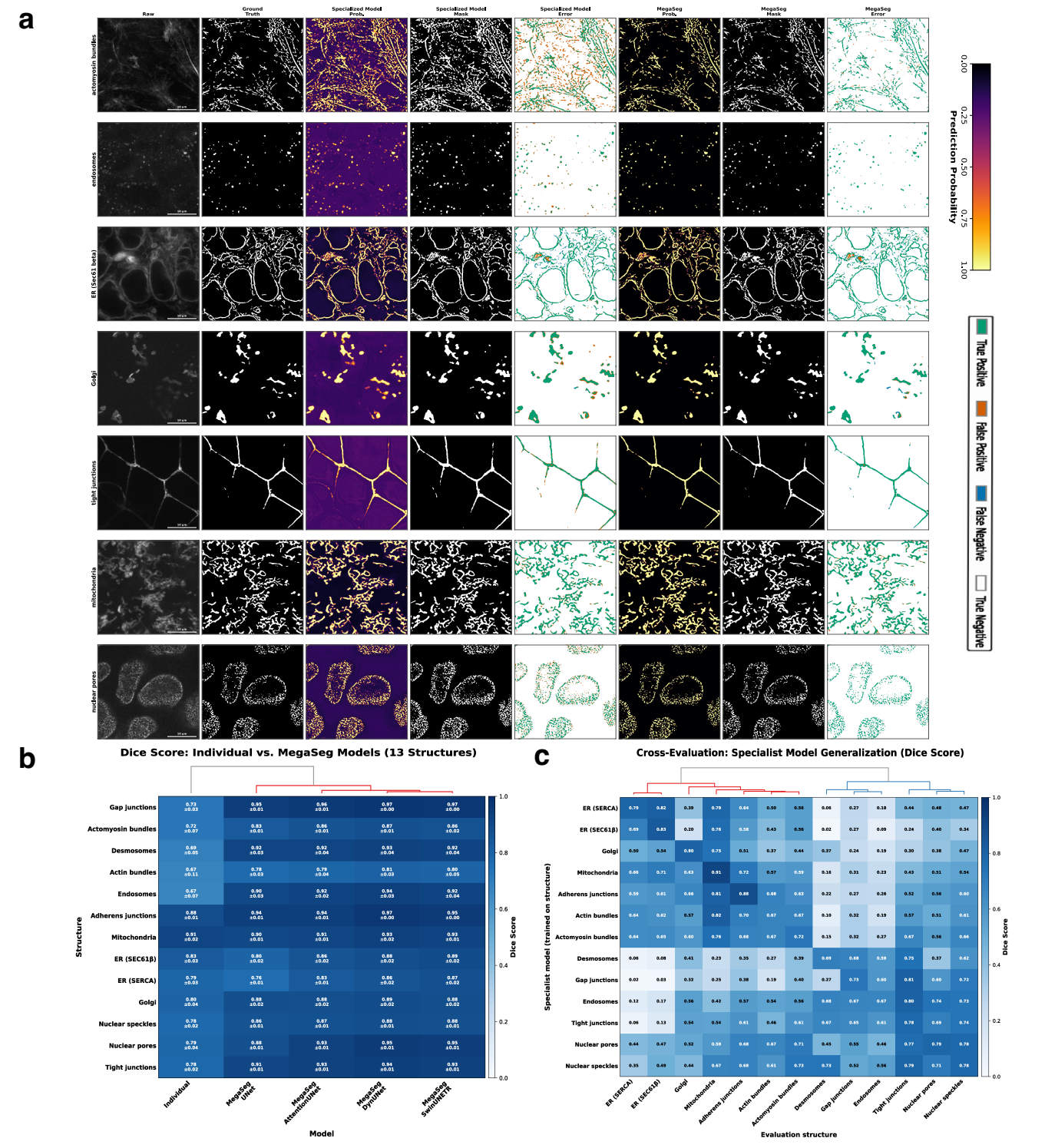

a
Raw
Ground Truth
Specialized Model Prob.
Specialized Model Mask
Specialized Model Error
MegaSeg Prob.
MegaSeg Mask
MegaSeg Error
actomyosin bundles
endosomes
ER (Sec61 beta)
Golgi
tight junctions
mitochondria
nuclear pores
Prediction Probability
0.00
0.25
0.50
0.75
1.00
True Positive
False Positive
False Negative
True Negative
b
Dice Score: Individual vs. MegaSeg Models (13 Structures)
Structure
Gap junctions
Actomyosin bundles
Desmosomes
Actin bundles
Endosomes
Adherens junctions
Mitochondria
ER (SEC61β)
ER (SERCA)
Golgi
Nuclear speckles
Nuclear pores
Tight junctions
Individual
MegaSeg UNet
MegaSeg AttentionUNet
MegaSeg DynUNet
MegaSeg SwinUNETR
Model
Dice Score
c
Cross-Evaluation: Specialist Model Generalization (Dice Score)
Specialist model (trained on structure)
ER (SERCA)
ER (SEC61β)
Golgi
Mitochondria
Adherens junctions
Actin bundles
Actomyosin bundles
Desmosomes
Gap junctions
Endosomes
Tight junctions
Nuclear pores
Nuclear speckles
Evaluation structure
Dice Score